Lecture 6: Stochastic algorithms

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Inverse problems in biological imaging
MSc Data Science and Artificial Intelligence
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L-smoothness: revision

Definition (L-smoothness)

Let $f \in \Gamma_0(\mathbb{R}^n)$ be differentiable. We say that f is an *L-smooth function* with constant $L \geq 0$ iff ∇f is L-Lipschitz continuous:

$$\exists L \geq 0 : \quad \forall w, z \in \mathbb{R}^n \quad \|\nabla f(w) - \nabla f(z)\| \leq L\|w - z\|.$$

Example: compute L-smoothness constant with $f(x) = \frac{1}{2}\|Ax - y\|^2$.

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Recall the definition of **operator norm**: if $B \in \mathbb{R}^{n_1 \times n_2}$ then:

$$\|B\| = \sup\left\{x \neq 0 : \frac{\|Bx\|}{\|x\|}\right\}$$

Note that by definition:

$$\frac{\|Bx\|}{\|x\|} \leq \|B\| \quad \forall x \neq 0 \quad \Rightarrow \quad \|Bx\| \leq \|B\|\|x\| \quad \forall x \neq 0.$$

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L is not dependent on w, z

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Y =Ax + n this is the inverse term. From here this tear came f(x)

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You saw previously that $\nabla f(x) = A^T(Ax - y)$. We look for $L > 0$ such that:

$$\|\nabla f(x_1) - \nabla f(x_2)\| = \|A^T(Ax_1 - y) - A^T(Ax_2 - y)\| = \|A^T A(x_1 - x_2)\|$$

$$\underbrace{\leq}_{\text{def. op. norm}} \quad \|A^T\|\|A(x_1 - x_2)\| \leq \|A^T\|\|A\|\|x_1 - x_2\| = \|A\|^2\|x_1 - x_2\|$$

So $L = \|A\|^2$.

Stochastic gradient descent

Motivations

Back to smooth optimisation problem:

$$\boxed{\min_{x \in \mathbb{R}^d} \; f(x)}$$

for differentiable, proper f with L-Lipschitz gradient ∇f.

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In the context of learning approaches, very often f relates to empirical-risk minimisation function. For a set of examples $\{y_1, \ldots, y_n\}$, with (typically) large $n \gg 1$:

$$\min_{x \in \mathbb{R}^d} \left\{ f(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x) = \frac{1}{2n} \sum_{i=1}^{n} \underbrace{\|A_i x - y_i\|^2}_{f_i(x)} \right\}$$

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sample no = n

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Recalling GD iteration, for $x_0 \in \mathbb{R}^d$ and $\tau \in (0, \frac{2}{L}]$, $k \geq 0$:

$$x_{k+1} = x_k - \tau\nabla f(x_k) = x_k - \frac{\tau}{n}\sum_{i=1}^{n} \nabla f_i(x_k) = x_k - \frac{\tau}{n}\sum_{i=1}^{n} A_i^T(A_i x_k - y_i)$$

Gradient evaluations may be costly (many matrix/vector products...)...

Motivations

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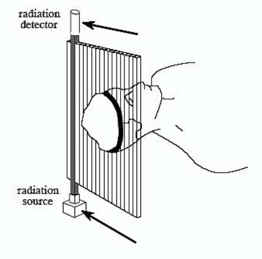
Gradient evaluations may be costly (many matrix/vector products. . .). . .

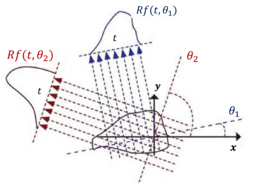
. . . can we use only one (or some) of the component(s) f_i to reduce computational costs?

A practical example: Computed Tomography (CT)

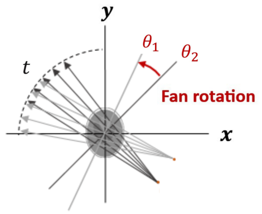
Idea: imaging technique that cross-sections the body by X-ray beams.



Measure changes along rays



Parallel beam geometry



Fan beam geometry

Goal: determine the attenuation coefficients of the X-rays caused by body absorption. Both the X-ray source and the detector array rotate, so as to measure attenuation coefficients from multiple angles.

$$\boxed{y = [\mathcal{R}x](\rho, \theta) = \int_{\mathbb{R}} x(\gamma_{\rho,\theta}(s))\, ds = Ax}$$

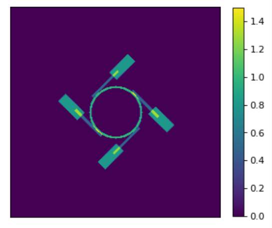
...+ noise

A practical example: Computed Tomography (CT)

Idea: decompose the CT operator $A \in \mathbb{R}^{N_\rho N_\theta \times N_{pix}}$ into $A_1, \ldots, A_{N_s} \in \mathbb{R}^{N_\rho n_\theta \times N_{pix}}$ where $n_\theta < N_\theta$ so that:

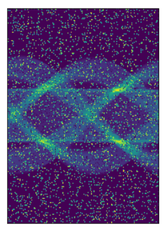
$$y_i = A_i x, \quad y_i = \chi_{Y_i}(y),$$

where $\chi_{Y_i}(\cdot)$ is the characteristic function of Y_i, a spatial sub-region in the sinogram.



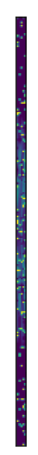
GT image $\hat{x}$

θ



ρ

Noisy sinogram y



y_i

stochastic approaches reduce the computational costs, especially in the presence of large imaging data!

Basic assumption

Use $\nabla f_i(x) \approx \nabla f(x)$?

Unbiased gradient estimator

Consider $i \in \{1, \ldots, n\}$ a random index selected **uniformly** at random (with probability $\frac{1}{n}$). Then:

$$\mathbb{E}_i[\nabla f_i(x_k)|x_k] = \sum_{i=1}^{n} \frac{1}{n} \nabla f_i(x_k) = \frac{1}{n} \sum_{i=1}^{n} \nabla f_i(x_k) = \nabla f(x_k)$$

"On average" the use of a single component ∇f_i provides an unbiased estimator of ∇f for uniformly sampled components i.

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Stochastic Gradient Descent (SGD): constant step-size

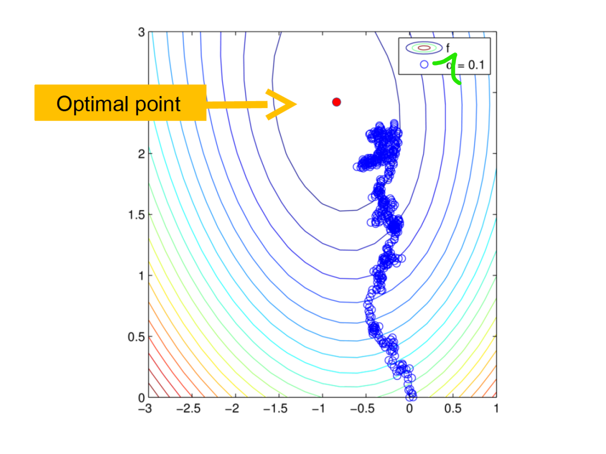
Input: $x_0 \in \mathbb{R}^d$ (initial guess), $\tau > 0$

Iterate for $k \geq 0$:

$$\text{sample uniformly } j \in \{1, \ldots, n\}$$
$$x_{k+1} = x_k - \tau \nabla f_i(x_k)$$

till **convergence**.

Intuitive behaviour



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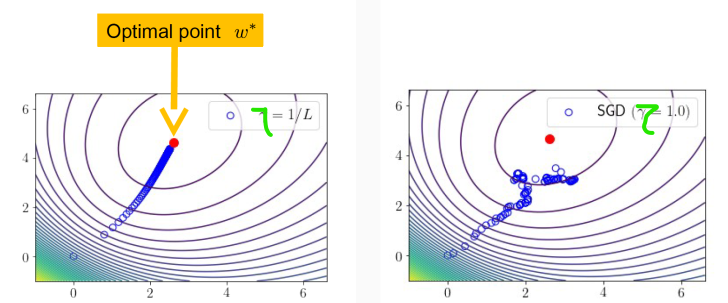


GD VS. SGD

Why does this happen? Need of assumptions!

Convergence of SGD (constant step-size)

Theorem: convergence of SGD (constant τ)

Let $x^* \in \arg\min f$. Let ∇f_i for $i = 1, \ldots, n$ be L_i-smooth and let $L_{max} := \max_i \; L_i$.
Then, the sequence (x_k) generated by SGD with $0 < \tau_k \equiv \tau < \frac{1}{2L_{max}}$ satisfies:

$$\mathbb{E}[f(\bar{x}_k) - f(x^*)] \leq \frac{C_1(x_0, x^*, \tau, L_{max})}{k} + \boxed{C_2(\tau, L_{max})\mathrm{Var}[\nabla f_i(x^*)]},$$

with $\bar{x}_k := \frac{1}{k}\sum_{j=0}^{k-1} x_j$.

- First term is similar in speed to what you get in non-stochastic GD ($O(1/k)$)
- Second term depends on

$$\mathrm{Var}[\nabla f_i(x^*)] = \mathbb{E}[\|\nabla f_i(x^*) - \nabla f(x^*)\|^2] = \mathbb{E}[\|\nabla f_i(x^*)\|^2]$$

- Possibly, very small step-size τ!

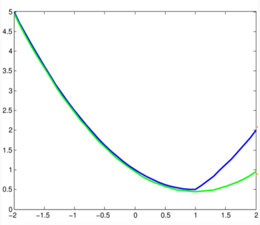
!

Improved condition for convergence: strong convexity

Defintion (strongly convex function)

Let $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be a proper and differentiable function. We say that f is **strongly convex** of parameter $\mu > 0$ (or μ-strongly convex) if:

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2, \quad \forall x, y \in \mathbb{R}^d.$$



Strongly convex functions have a quadratic lower bound at every point.

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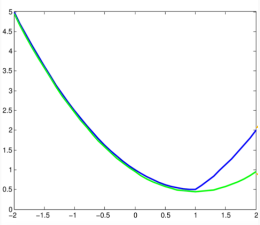


Strongly convex functions have a <u>quadratic lower bound</u> at every point.

Important remark: strong convexity $\Rightarrow$ strict convexity $\Rightarrow$ convexity

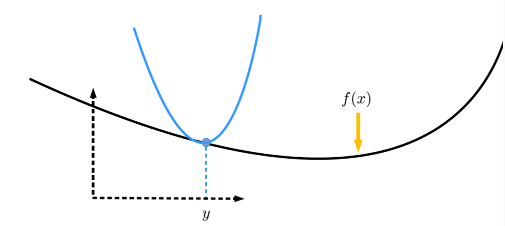
Exercise: $f(x) = x^4$ is a strictly convex function which is not strongly convex.

From lower to upper bounds via L-Lipschitz smoothness

Proposition (quadratic upper bound)

Let $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be a proper and differentiable function with L-Lipschitz gradient. Then:

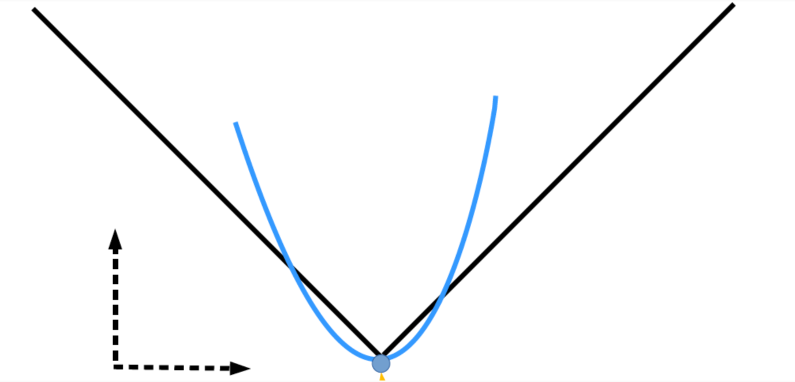
$$f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2}\|x - y\|^2, \quad \forall x, y \in \mathbb{R}^d.$$



L-smooth functions have a quadratic **upper** bound at every point.

Test this property, convex counter-example

$f(x) = \|x\|_1$, convex.



Can't define a quadratic upper bound in $x = 0$

Convergence result

Theorem (Convergence of SGD, strongly cvx. case, constant τ)

If f is μ-strongly convex and all ∇f_i are L_i-Lipschitz continuous for $i = 1, \ldots, n$, denoting by $x^* = \arg\min\ f(x)$ and defining:

$$L_{max} := \max_{i=1,\ldots,n} L_i, \qquad \mathrm{Var}[\nabla f_j(x^*)] := \mathbb{E}\left(\|\nabla f_j(x^*)\|^2\right),$$

<u>the iterates</u> (x_k) of SGD with $\tau \leq \frac{1}{2L_{max}}$ satisfy:

$$\mathbb{E}\left(\|x_k - x^*\|^2\right) \leq (1-\tau\mu)^k \underbrace{\|x_0 - x^*\|^2}_{C_1} + \underbrace{\frac{2\tau}{\mu}}_{C_2} \mathrm{Var}[\nabla f_j(x^*)]$$

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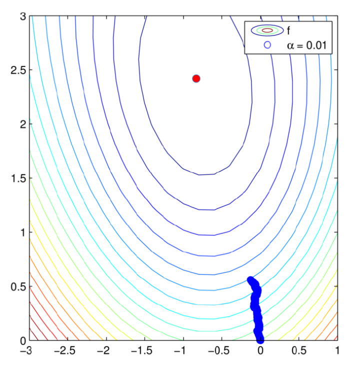
Note that in order to have $\mathbb{E}\left(\|x_k - x^*\|^2\right) \to 0$, we need:

- To get $(1-\tau\mu)^k \to 0$ fast, we hope $1 - \tau\mu \approx 0$, i.e. $\tau = 1/\mu \gg 1$
- To get $\frac{2\tau}{\mu}\mathrm{Var}[\nabla f_j(x^*)] \to 0$, I need τ small

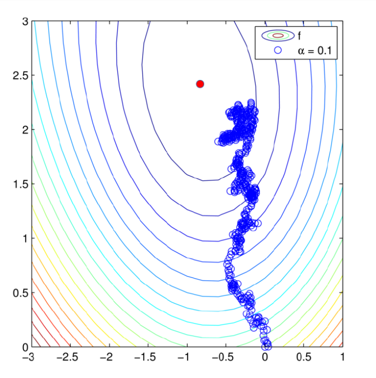
Not really compatible. . .

Small $\tau \approx 0 \rightarrow$ small variance



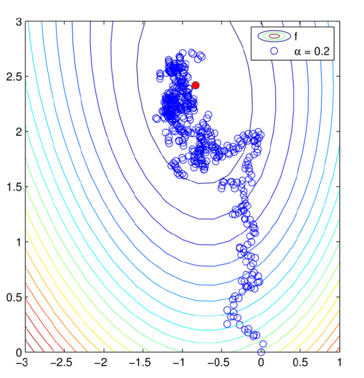
Convergence of SGD for fixed step-size

$\tau = 0.1$

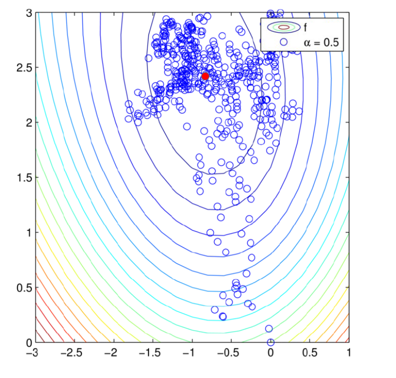


Convergence of SGD for fixed step-size

$\tau = 0.2$



$\tau = 0.5 \rightarrow$ large variance



Idea: choose large τ_k in early iterations to get close to the minimiser and then reduce it to reduce the variance.

SGD with varying step-sizes

$\tau_k = \frac{1}{k}$

Stochastic Gradient Descent (SGD): decaying step-size

Input: $x_0 \in \mathbb{R}^d$ (initial guess), (τ_k) s.t. $\sum_{k=1}^{\infty} \tau_k = +\infty$, $\tau_k \to 0$

Iterate for $k \geq 0$:

$$\text{sample uniformly } j \in \{1, \ldots, n\}$$

$$x_{k+1} = x_k - \tau_k \nabla f_j(x_k)$$

till **convergence**.

Theorem (Convergence of SGD, decaying step-sizes)

If f is μ-strongly convex, all ∇f_i are L_i-Lipschitz continuous, let $x^* = \arg\min f(x)$ and:

$$L_{max} := \max_{i=1,\ldots,n} L_i, \quad \mathrm{Var}[\nabla f_j(x^*)] := \mathbb{E}\left(\|\nabla f_j(x^*)\|^2\right), \quad \kappa := \lceil L_{max}/\mu \rceil.$$

for the following choice of step-sizes:

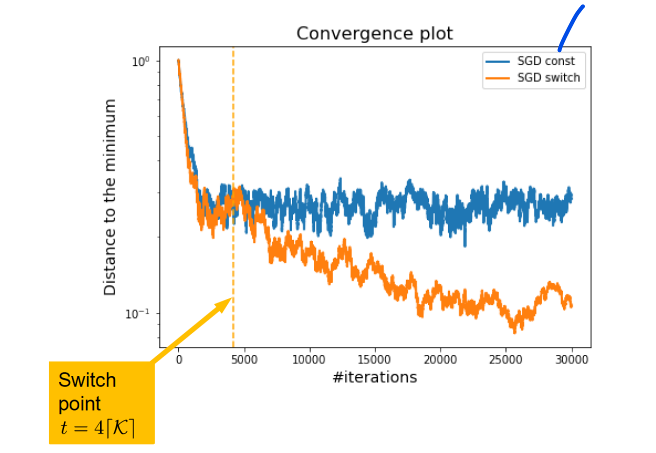
$$\tau_k = \begin{cases} \frac{1}{2L_{max}} & \text{if } k \leq 4\kappa \\ \frac{2k+1}{(k+1)^2\mu} & \text{if } k > 4\kappa \end{cases}$$

the following convergence result holds true for $k \geq 4\kappa$:

$$\mathbb{E}\left(\|x_k - x^*\|^2\right) \leq \frac{8\sigma^2}{\mu^2 k} + \frac{16\kappa}{e^2 k^2}\|x_0 - x^*\|^2$$

Note: RHS$\to$ 0. Note that the decreasing step-size is $\approx O\left(\frac{1}{k+1}\right)$. Practically, often a slowest decay $\tau_k = \frac{C}{\sqrt{k+1}}$, for tuned $C > 0$ is chosen.

Convergence comparisons



How to avoid oscillations?

Stochastic gradient descent with averaging

SGD with averaging

SGD with varying step-size and (late) averaging

Input: $x_0 \in \mathbb{R}^d$ (initial guess), (τ_k) s.t. $\sum_{k=1}^{\infty} \tau_k = +\infty$, $\tau_k \to 0$, $s_0 \in \mathbb{N}$

Iteration index of our choice.

Iterate for $k \geq 0$:

sample uniformly $j \in \{1, \ldots, n\}$

$$x_{k+1} = x_k - \tau_k \nabla f_j(x_k)$$

if $k > s_0$

$$\bar{x} := \frac{1}{k - s_0} \sum_{i=s_0}^{k} x_i$$

else

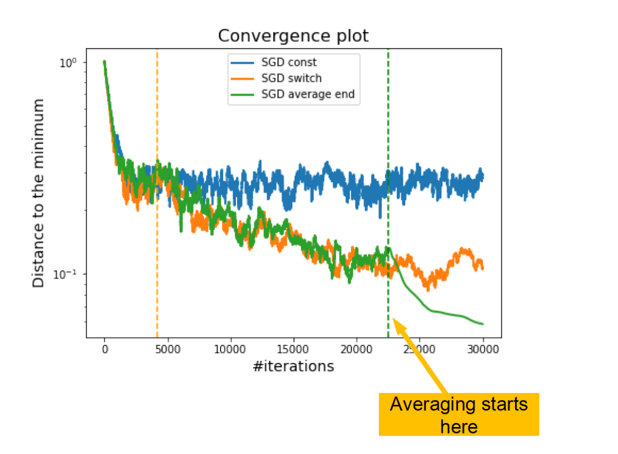
$$\bar{x} = x_k$$

$$x_k = \bar{x}$$

till **convergence**.

Strategy employed in: Polyak, Juditsky, *Acceleration of stochastic approximation by averaging*, SIAM Journal on Control and Optimization, 1992.

Further convergence comparisons

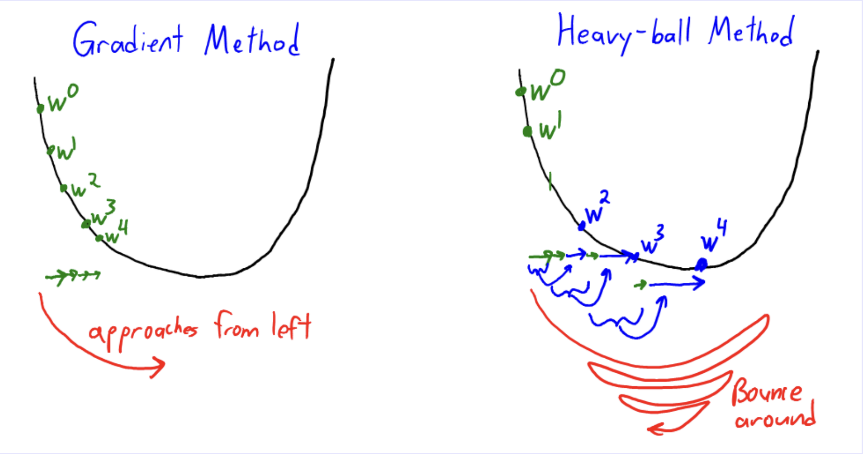


Acceleration strategies

We have seen already the idea of "inertia" (à la FISTA) to improve convergence speed.

Heavy ball acceleration (deterministic):

$$x_{k+1} = x_k - \tau \nabla f(x_k) + \beta_k (x_k - x_{k-1}), \quad \beta_k \in (0, 1)$$



Back to GD with momentum

We have seen already the idea of "inertia" (à la FISTA) to improve convergence speed.

Heavy ball acceleration (deterministic):

$$x_{k+1} = x_k - \tau \nabla f(x_k) + \beta_k (x_k - x_{k-1}), \quad \beta_k \in (0, 1)$$

Momentum method (stochastic):

$$x_{k+1} = x_k - \tau \nabla f_{i_k}(x_k) + \beta_k (x_k - x_{k-1}), \quad \beta_k \in (0, 1)$$

with $i_k \in \{1, \dots, n\}$ drawn uniformly.

Convergence of momentum method

Theorem (Convergence of SGD with momentum)

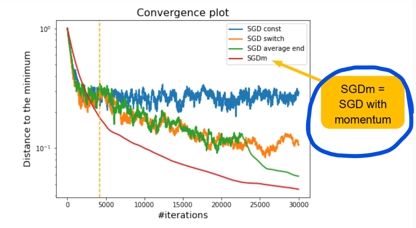
By taking the parameters as:

$$\tau_k = \frac{2\eta}{k+3}, \quad \beta_k = \frac{k}{k+1}, \quad \eta \leq \frac{1}{4L_{max}},$$

then the following result holds for SGD with momentum:

$$\mathbb{E}[f(x_k) - f(x^*)] \leq \frac{\|x_0 - x^*\|^2}{\eta(k+1)} + 2\eta \mathrm{Var}[\nabla f_i(x^*)]$$

Not **faster**, but **stronger** as convergence of sequence $f(x_k)$ (not of average $f(\bar{x}_k)$)!



Nods on variants and non-smooth alternatives

Variants of SGD

- **Averaging**: close relations ($\approx$ equivalence) with SGD with momentum approaches.
- **AdaGrad**, **RMSProp** [1]: SGD with adaptive (component-dependent) learning rate.
- **ADAM** [2]: adaptive algorithm using both estimation of first and second moment of the gradients.

Generalisation to non-smooth problems ($\approx$proximal algorithms?)

$$\min_{x\in\mathbb{R}^d} \frac{1}{n}\sum_{i=1}^{n} f_i(x) + g(x)$$

[1] Duchi, Hazan, Singer, '11, Hinton, '12
[2] Kingma, Ba, '14

Stochastic proximal gradient algorithm

$$\min_{x \in \mathbb{R}^d} \; F(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x) + g(x)$$

- f_i is differentiable, has L-Lipschitz gradient
- g is proper, convex and l.s.c
- $x^* \in \arg\min \; F$

Stochastic proximal gradient descent (constant τ)

$x_0 \in \mathbb{R}^n$, $\tau < \frac{1}{4L_{max}}$

$$i_k \in \{1, \dots, n\} \quad \text{with probability } \frac{1}{n}$$

$$x_{k+1} = \text{prox}_{\tau g}(x_k - \tau \nabla f_{i_k}(x_k))$$

Convergence of stochastic proximal gradient algorithm

Theorem (convergence of SPGD with constant τ)

Let (x_k) be the sequence generated by SPGD with constant step-size $\tau < \frac{1}{4L_{max}}$. Then for all $k \geq 1$:

$$\mathbb{E}[F(\bar{x}_k) - F(x^*))] \leq \frac{\|x_0 - x^*\|^2 + 2\tau(F(x_0) - F(x^*))}{2(1 - 4\tau L_{max})\tau k} + \frac{2\tau \mathrm{Var}[\nabla f_i(x^*)]}{(1 - 4\tau L_{max})},$$
$$= \frac{C_1(x_0, x^*, \tau, L_{max})}{k} + C_2(\tau, L_{max})\mathrm{Var}[\nabla f_i(x^*)]$$

with $\bar{x}_k = \frac{1}{k}\sum_{j=0}^{k-1} x_j$.

Again, "average" result.

- In the case of strong convexity, the rate is of the form $(1 - \mu\tau)^k$
- Acceleration (momentum) in non-smooth stochastic contexts is very tricky, not very clear how to perform it
- Here stochastic gradients... stochastic proximal operators?

See A. Khaled et al., 2020 for more variants.

Questions?

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